

PROJECT REPORT

Generating text, image and audio together

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ABSTRACT

We built a system that produces three kinds of content — a written passage, an image, and a spoken narration — from a single short prompt, with each modality generated independently by its own model. The generator runs entirely on one provider’s free tier and returns each modality in under two seconds. On text quality it sits within a few points of the large commercial models; on image aesthetics and audio it trails them, as an eight-billion-parameter open model should be expected to.

The substantive contribution is not the generator but the evaluation. Because the three models never see one another’s output, nothing guarantees the three results describe the same thing, and no conventional metric asks whether they do — each compares one output against the prompt in isolation. We therefore developed a cross-modal coherence score, calibrated a set of per-modality quality grades against an LLM judge, and compared four candidate scoring methods on 53 prompts.

Two findings are firm. Plain averaging of cosine similarities across different encoders correlates negatively with judged quality and is unusable, because the similarities are not commensurable: one pair is systematically larger than the others on every single prompt for reasons internal to the encoder. And one of our three quality features carries no signal at all, which we can now attribute to it duplicating another feature at a rank correlation of 0.902. A third result — whether the quality-weighted coherence score predicts judged quality — is contested between two implementations in our own team and is reported here as unresolved rather than settled in favour of the more flattering number.

1 Introduction

Generative models have become reliable at producing one kind of output well. Producing three at once from a single instruction is a different problem, and judging the result is a harder one still. This report describes both, and is honest about which parts we consider finished.

The system takes a short description — an owl librarian wearing glasses — and returns a paragraph about it, an image of it, and an audio narration of that paragraph. Three separate models do the work, each from the same prompt, and their outputs are combined only at the end. This separation is deliberate. It keeps each modality free to use whichever model suits it, it mirrors the multi-agent designs of recent research systems [14, 15], and it makes the architecture straightforwardly extensible.

It also creates the problem that came to dominate the project. If three generators work from the same prompt but never consult one another, nothing enforces that their outputs are about the same thing. A system can return a librarian owl in the text, a plain barn owl with no glasses in the image, and narration about something else, and every standard metric will report success — because every standard metric compares a single output against the prompt. BERTScore only sees text. CLIP only compares image to text. CLAP only compares audio to text. None of them look across the outputs.

Measuring that agreement turns out to be considerably harder than it appears, for a reason that is a property of the measuring instruments rather than of our system. That is the subject of Sections 5 to 7.

2 Objectives

The project set six objectives, of which four are met and two are partly met.

Objective	Status
Build a working generator producing text, image and audio from one prompt	Met
Select models by benchmarking rather than preference	Met
Place the three modality scores on a common 0–1 scale	Met
Develop a measure of cross-modal coherence	Met; its validity is in question (§7)
Calibrate the evaluation and identify the best scoring method	Partly met; result contested (§6.4)
Validate the evaluation against human judgement	Not started

3 Related work

The closest system to ours is MM-StoryAgent [14], which assembles narrated storybook videos from cooperating agents — story agents that brainstorm, image agents that hold characters consistent across scenes, and three separate audio channels. Its reviewer loop raised CLIP alignment from 0.297 to 0.316, and its absolute CLIP score is almost identical to ours, which is a useful sanity check on our own numbers. MAGUS [15] shares our philosophy of coordinating existing models rather than retraining one, but its generate-score-prune loop proved so expensive that its authors could not fully tune it. NExT-GPT [16] is the contrast case: a single trained any-to-any model requiring GPUs, where we make free API calls from a laptop.

Two of these papers are candid about the same gap we hit. PresentAgent [18] states plainly that its metrics assess visual, audio and text dimensions independently rather than their cross-modal coherence. That is precisely the limitation this project set out to address.

On the evaluation side, our metrics are standard and traceable: CLIPScore [5] for image alignment, the LAION predictor for aesthetics, BERTScore [9] for text, Whisper [8] for transcription, CLAP [7] for audio alignment, and DNSMOS [11] for speech quality. We also read an audit of the LAION aesthetic predictor [13] which finds it culturally biased towards Western and Japanese subjects, because its training ratings came largely from English-speaking Western annotators; we therefore treat the aesthetic column as indicative rather than as a measure of quality.

The papers that mattered most arrived late, after our first coherence result looked wrong. Lu et al. [19] score text–image pairs on two channels at once — an embedding metric and an LLM judge on a rubric — and validate both against human annotators; this is the direct source of our judge-and-regression design. Dhimoila et al. [21] introduce the Iso-Energy Assumption and show that unimodal components of an embedding act as modality-specific bias and fully account for the modality gap; this explains our central negative result. Nandakumar et al. [24], from biometrics rather than multimodal generation, supplied the principled score-fusion alternative to our hand-tuned formula.

4 System design

4.1 Architecture

The system is a layered pipeline orchestrated with LangGraph behind a Streamlit interface. A supervisor initialises a shared state object; a router selects the agent for the requested modality; each agent then runs the same six-stage sequence of analyzer, retriever, generator, optimizer, reviewer and exporter, with a retry edge from reviewer back to optimizer when the score falls below a threshold, capped so a difficult prompt cannot loop indefinitely.

4.2 Provider selection

Two approaches were weighed. Using a specialist provider per modality buys best-in-class models but requires a separate account, key and billing arrangement for each — more than ten integrations. Using one provider for all three requires one of each.

The decision was settled by what happened when we tried the first approach. Of four text providers, three worked. Of four image providers, one worked. Of seven audio providers, none produced a usable file: two were speech-to-text services we had misidentified, three were blocked behind payment, one failed on a Windows toolchain error, and one worked only as a local fallback we never integrated. A modality is not genuinely available to a student project until some provider will serve it on a free tier without a card on file, and in May 2026 we had none for audio.

Cloudflare Workers AI resolved this by hosting all three modalities behind a single account and key.

Modality	Model	Neurons	Latency
Text	Llama 3.1 8B Instruct	26	1.85 s
Image	FLUX.1 [schnell]	58	1.53 s
Audio	MeloTTS	5	1.68 s

The free tier allows 10,000 neurons daily, Cloudflare’s own compute unit; a full run over 56 prompts costs about 4,350, permitting roughly two runs a day. That ceiling shaped the experimental design more than any methodological choice did, and it is the main argument for a funded API in any continuation of this work.

Three undocumented platform behaviours are worth recording because each would otherwise corrupt results silently. Roughly a quarter of MeloTTS calls return HTTP 500, confirmed as the provider’s fault by sending an identical request eight times and observing two failures with no relation to the input; a three-attempt retry resolves it. The image endpoint returns JPEG bytes in files named .png. And the audio is sampled at 44.1 kHz rather than the 48 kHz we assumed, which matters because each scorer expects a different rate and resampling incorrectly yields plausible but wrong numbers rather than an error.

5 Evaluation methodology

Evaluation was built in layers, each addressing a shortcoming in the one before.

5.1 Per-modality quality grades

The base metrics occupy incompatible scales: BERTScore runs 0 to 1, aesthetics 0 to 10, MOS 1 to 5, and WER is better when lower. Each is first mapped onto 0–1 against a fixed external standard — the ITU scale for MOS, the LAION scale for aesthetics — rather than against the spread of our own data, so that a grade means the same thing across runs.

Within a modality, two related sub-scores are combined by a formula that rewards both being high and penalises imbalance:

$$\text{quality_pair}(a, b) = \text{avg}(a, b) - \lambda \times \text{variance}(a, b), \quad \lambda = 0.5$$

$Q_{\text{text}} = \text{BERTScore}(\text{generated_text}, \text{prompt})$

$Q_{\text{image}} = \text{quality_pair}(\text{CLIP_score}, \text{aesthetic_score})$

$Q_{\text{audio}} = \text{quality_pair}(\text{semantic_score}, 1 - \text{WER})$

The reasoning behind the penalty is that a strong CLIP score should not be permitted to compensate for an unattractive image; a good pair should be good on both channels.

5.2 Replacing CLAP

CLAP is the natural audio analogue of CLIP and we intended to use it. On our data it returned exactly zero for 25 of 53 prompts — 47 per cent — with a mean of 0.082. The explanation is a mismatch of purpose rather than a defect: laion/clap-htsat-unfused was trained on environmental sound and music, and our audio is text-to-speech narration. It listens for sound events, finds none, and reports no alignment irrespective of the words spoken.

We replaced it with a two-step semantic measure — transcribe with Whisper, then compute BERTScore between transcript and prompt — which yields usable signal on every prompt at a mean of 0.855. CLAP is retained in the pipeline solely as a comparison column documenting why it was dropped.

5.3 Calibration against a judge

Weighting the three grades equally was a guess. To replace it, Gemini is shown the prompt, the generated text, the generated image and the audio transcript, and returns an overall score and per-modality scores from 1 to 10 with a one-sentence justification. All 53 prompts were rated; overall mean 6.81, with text at 8.11, image at 8.25 and audio at 4.66.

A linear regression of the judge’s overall score on the three standardised quality features gives normalised weights of 0.287 for text, 0.520 for image and 0.193 for audio. Image quality dominates, consistent with the judge’s own per-modality pattern.

5.4 Cross-modal coherence

The central idea is to encode all three outputs into one shared vector space and measure the cosine similarity of each pair, then average and penalise disagreement:

$s1 = \text{sim}(\text{text}, \text{image}) \quad s2 = \text{sim}(\text{text}, \text{audio}) \quad s3 = \text{sim}(\text{image}, \text{audio})$

$\text{TSAS} = (s1 + s2 + s3) / 3$

$\text{Coherence} = \text{TSAS} - \lambda \times \text{variance}(s1, s2, s3)$

Two implementations were built. The first runs all three modalities through ImageBind [6], which maps six modalities into a single 1024-dimensional space; audio is resampled to 16 kHz, converted to a 128-bin log-mel spectrogram and padded to 204 frames, because ImageBind’s audio encoder treats spectrograms as images. The second uses specialist encoders — CLIP for text–image, CLAP for text–audio, and for image–audio a Whisper transcript compared to the image by CLIP, which is a comparison CLIP was actually trained to make.

A quality-weighted variant, WTSAS, scales each similarity by the average learned quality of the two modalities it connects, so that agreement between two strong outputs counts for more than agreement between a strong and a weak one.

5.5 Alternative scoring methods

To have principled alternatives rather than only hand-tuned formulas, we specified three methods from the statistics literature. A Naive Bayes posterior bins the judge’s score into Good, Medium and Bad, fits a Gaussian per feature per class, and takes the continuous posterior probability of Good. A likelihood ratio

score, after Nandakumar et al. [24], fits one Gaussian to matched triples and another to mismatched ones and sums the log ratios; since summing log-likelihood-ratios over independent features is exactly Naive Bayes, these two are views of the same mathematics. A Mamdani fuzzy system was specified on the argument that probability models randomness while fuzzy logic models vagueness; it has not been implemented, and we note that rather than omitting it.

6 Results

6.1 Generation quality

Measure	Ours	Claude	ChatGPT	Gemini
Text — BERTScore	0.844	0.851	0.832	0.796
Text — ROUGE-L	0.056	0.078	0.071	0.068
Text — Readability	0.948	1.85	1.72	1.68
Image — CLIP	0.31	0.312	0.305	0.298
Image — Aesthetic (/10)	5.80	7.20	7.05	6.90
Audio — MOS (/5)	3.37	4.10	4.05	4.20
Audio — WER	0.095	0.072	0.085	0.068

On text the system is within a few points of the commercial models. On image aesthetics it is roughly 1.2 points behind on a ten-point scale, and on audio it trails on both measures. Three figures the commercial vendors do not publish are ours to report: response under two seconds per modality, a complete run inside a free daily quota, and roughly 90 per cent accuracy on an automated image fact-check in which one model writes yes/no questions from the prompt and a vision model answers them from the image. Forty-two of 56 images passed every question. Several failures were not the generator’s fault but the method’s — questions such as is music playing cannot be answered from a still image at all.

6.2 Which features carry signal

Before comparing methods, the raw features are worth examining individually.

Feature	Spearman	p	Interpretation
Q_text	+0.316	0.021	significant, usable
Q_image	+0.297	0.031	significant, usable
Q_audio	−0.006	0.969	no relationship

Q_audio is not weakly correlated with judged quality; it is uncorrelated. This shapes every result that follows, because any method obliged to weight all three features equally is carrying a feature that contributes nothing.

The reason emerged only when the larger dataset arrived, and it is a design fault rather than a sampling artefact. In the 500-prompt data, bertscore and semantic correlate at Spearman 0.902. This follows directly from the pipeline: the TTS reads the generated text, Whisper transcribes it accurately, so BERTScore of the transcript against the prompt is very nearly BERTScore of the generated text against the prompt. Half of Q_audio re-measures Q_text. No amount of additional data will repair that.

6.3 Method comparison

Each method was run on the same 53 prompts and correlated against the judge’s overall score. We report rank-based correlations, since at this sample size Pearson is unduly sensitive to individual outliers.

Method	Spearman	Kendall	Pearson
WTSAS (quality-weighted)	0.451	0.350	0.399
Likelihood ratio	0.312	0.240	0.259
Naive Bayes	0.296	0.231	0.241
TSAS (plain averaging)	−0.077	−0.056	−0.085

The two probabilistic methods land close together, as their shared mathematics predicts, and both improve when Q_{audio} is excluded — Naive Bayes moves from 0.211 to 0.296. Plain TSAS is negative, and that result is reproduced in both of our independent implementations.

7 Discussion

7.1 Why plain averaging fails

The most robust finding in this project is negative, and it is worth stating precisely because it generalises beyond our system.

Pair	Mean
s1 — text ↔ image	0.362
s2 — text ↔ audio	0.106
s3 — image ↔ audio	0.116
TSAS	0.195

The text–image similarity is roughly three times either audio pair, and it is larger on every one of the 53 prompts — not on a majority. A gap that uniform cannot be a property of the content; it appears whether the prompt describes a librarian owl or a taco lifting weights.

The explanation is structural. ImageBind’s audio encoder sits further from its text and image encoders in the shared space than those two sit from each other, independently of whether the audio matches. This is the modality gap, and Dhimoila et al. [21] show that it is fully accounted for by unimodal components acting as modality-specific bias — components that can be removed without harming performance. Since the coherence score is computed directly on the three similarities, and one is systematically largest, the variance penalty partly measures a fixed property of the encoder rather than the agreement of a particular triple. A perfectly matched set is penalised for the same reason a mismatched one is. Increasing λ worsens this: at $\lambda = 2.0$ the mean score falls to about half of mean TSAS, almost entirely through a structural penalty.

The implication for anyone building a similar metric is that raw cosine similarities from different encoder pairs are not commensurable and must not be averaged directly. Each needs reading against its own baseline — which is why a mismatched-pair floor, obtained by shuffling prompts against outputs, is the correction we are now building.

7.2 Why the probabilistic methods underperform

Setting aside the disputed row, both probabilistic methods reach the low 0.3s, and there are three specific reasons rather than a general one.

First, binning discards exactly what the evaluation rewards. Naive Bayes requires the continuous judge score to be split into classes before it can fit anything, but Spearman rewards correct fine-grained ordering, and a 7.1 and a 7.9 may fall in the same class. A regression on the raw continuous target never loses that ordering.

Second, the smallest class holds four examples. A Gaussian fitted from four points describes noise rather than structure. The instability is visible directly: the Naive Bayes correlation moved between 0.05 and 0.30 purely as a function of which features were included, and a Good-versus-Bad split, which discards the Medium class, collapses to 0.165.

Third, multiplying per-feature likelihoods offers no mechanism for discounting an uninformative feature, so including `Q_audio` actively degrades the score. A regression simply learns a low weight for it. None of these three is an argument against probabilistic scoring as such; two are consequences of sample size and one of feature quality.

7.3 What the judge can and cannot stand in for

Every correlation in this report is measured against an LLM judge acting as a proxy for a human evaluator. That was a defensible shortcut for rapid iteration, and the judge’s per-modality pattern is at least plausible — it rates image highly and audio poorly, matching our own measurements and our subjective impression. But Lu et al. [19], the paper from which we took the two-channel design, validate their embedding metric and their LLM judge both against human annotators, reporting inter-annotator agreement. We have not done the equivalent, so we cannot currently distinguish a metric that tracks human judgement from one that has merely learned to track Gemini.

8 Limitations

Four limitations bear materially on how the results above should be read.

Sample size. Fifty-three prompts, with a smallest class of four. The learned weights have a leave-one-out R^2 of 0.04, which is close to no predictive power, and they should be treated as indicative rather than as a finding. A 500-prompt extension is under way but currently lacks the judge ratings and the ImageBind similarities needed to repeat the comparison.

Reproducibility. Generation sets no temperature, so every model runs at its default and the same prompt produces different output, and different scores, on each run. Each prompt is also scored from a single generation. Fixing the temperature and reporting three to five runs with a standard deviation would substantially strengthen every number here.

Borrowed thresholds. The score thresholds are taken from external conventions rather than calibrated on our own labelled data, so they are reasonable defaults rather than tuned parameters.

No human validation. As set out in §7.3, nothing here has been checked against a person’s judgement.

A fifth issue is narrower but worth recording, since it affects comparison between our two datasets: `wer_inv` is computed against a different reference text in each. In the 53-prompt set it is $1 - \text{WER}(\text{generated_text}, \text{transcript})$, ranging 0.17 to 0.25; in the 500-prompt set it is $1 - \text{WER}(\text{prompt}, \text{transcript})$, ranging 0.88 to 0.93. These are different quantities sharing a column name, and `Q_audio` consequently has a different baseline in each. The interim workaround is z-score normalisation fitted on the labelled set; the proper fix is to recompute one against the other’s reference.

9 Conclusion and future work

We built a multi-agent generator that produces text, image and audio from a single prompt on one provider's free tier, benchmarked the provider choice against alternatives rather than assuming it, and developed a layered evaluation running from raw metrics through calibrated quality grades to cross-modal coherence.

The clearest contribution is a negative result with a mechanism attached. Averaging cosine similarities across different encoder pairs does not measure coherence, because those similarities are not commensurable: one pair is systematically larger on every prompt for reasons internal to the encoder rather than the data, and a variance penalty computed over them measures the instrument rather than the outputs. This is consistent with recent work on the geometry of the modality gap [21], and it means any metric of this family needs a per-pair baseline before the numbers can be combined.

The second contribution is diagnostic. One of our three quality features carries no signal, and we can now say why: it duplicates another feature at a rank correlation of 0.902, because the narration reads the generated text and the transcription recovers it accurately. This is a design fault in the feature, not a shortage of data.

Four things follow, in order of priority. The WTSAS discrepancy in §6.4 must be resolved by running both implementations on identical inputs, since nothing downstream can be defended until it is. The audio feature needs replacing with one that captures something the text does not — prosody or expressiveness, as EmergentTTS-Eval [12] argues WER and MOS both miss. A batch of twenty to thirty triples should be hand-scored for whether the three outputs match, and every candidate method ranked against those labels rather than against a language model. And the 500-prompt set needs its judge ratings and ImageBind similarities before it can serve the purpose for which it was collected.

Beyond those: the fuzzy method remains specified but unimplemented, the mismatched-pair baseline is designed but not yet a standard column, and fixing the generation temperature would make the whole benchmark reproducible. We consider the generator finished and the evaluation genuinely in progress, and have written this report to reflect that distinction rather than to obscure it.

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